# Tzu-Ming Harry Hsu

徐子旻

**Physical AI Researcher** h@stmharry.io
*World Models, Embodied AI, and Robot Learning* *+1 617-803-7785 / +886 928-494-198*

- Member of Technical Staff at Moonlake AI building simulation infrastructure for robotics, autonomy, and physical AI.
- MIT-trained Ph.D. and Google Federated Learning core contributor, with YFLOP-scale experimentation infrastructure and open-source research artifacts.
- Founder-operator with capital-raising, team-building, and production ML experience across data, training, evaluation, and scalable inference.
- International Physics Olympiad world champion, bringing first-principles problem solving to ambitious technical systems.

## *Education*

**Massachusetts Institute of Technology** Jun 2020 – Jun 2022
*Ph.D., Computer Science* *Cambridge, MA*

- *Advisor*: Peter Szolovits
- *Research areas*: Computer vision, federated learning, machine learning for healthcare

**Massachusetts Institute of Technology** Sep 2017 – May 2020
*S.M., Electrical Engineering and Computer Science* *Cambridge, MA*

- *Advisors*: Peter Szolovits, Fadel Adib
- *Research areas*: Computer vision, machine learning, wireless signal, signal processing

**National Taiwan University** Sep 2011 – Jun 2016
*B.S., Physics; B.S.E., Electrical Engineering* *Taipei, Taiwan*

- *Class rank*: 1/190
- *GPA*: 3.99 / 4.00

## *Professional Experience*

**Moonlake AI** Jul 2026 – Present
*Member of Technical Staff* *San Francisco, CA*

Build simulation and evaluation infrastructure for robotics, autonomy, and physical AI, spanning world generation, robot-learning environments, and policy evaluation.

**Dentscape** Apr 2025 – Apr 2026
*Machine Learning & Software Engineering Consultant* *San Francisco, CA / Taipei, Taiwan*

Built data, training, and evaluation systems for AI-assisted dental design, combining geometric ML, pre-labeling workflows, preference modeling, and production deployment improvements.

- Reorganized a fragmented ~10 TB dental dataset into a standardized ~4 TB ML-ready registry, reducing raw-data access from ~3 days to minutes.
- Rebuilt pre-labeling and labeling loops for 100-200 case batches, reducing turnaround from ~4 weeks to ~1 week and improving experiment velocity.
- Implemented bilevel/meta-learning for personalized crown generation and improved preference fit by ~20% while keeping the system production-facing.
- Optimized segmentation and crown-generation services for ~110% aggregate throughput improvement while reducing infrastructure cost by ~50%.

**Clarq AI** Dec 2024 – Jun 2026
*Co-founder & CTO* *Palo Alto, CA / Taipei, Taiwan*

Helped reposition Clarq from a hardware product into a force-intelligence platform, framing a long-term moat around force-time data, evaluation loops, model-driven product decisions, and fundraising.

- Co-led technical positioning around a three-layer strategy: hardware wedge, SaaS profit engine, and long-term data/model moat.
- Drove US-facing investor outreach and progressed pre-seed conversations toward near-commit stage with angels.
- Supported Startup Island Taiwan Silicon Valley program participation and international expansion narrative.
- Aligned product roadmap, data strategy, and investor-facing story across Taiwan hardware operations and US capital market expectations.

**CodeGreen Labs PBC** Sep 2023 – Nov 2024
*Co-founder & CTO* *Boston, MA / Taipei, Taiwan*

Led a 10-person engineering organization delivering auditable data systems for enterprise partners, reaching projected $1MM annual revenue while hardening integration, compliance, and delivery operations.

- Reached projected annual revenue of $1MM USD by Q2 2024 with a 10-person team.
- Established partnerships with Microsoft, Global Blockchain Business Council, Gold Standard, SDG Data Alliance, and PVBLIC.
- Built a cross-functional team across product, partnerships, and engineering to address complex data, auditability, and compliance challenges.
- Designed and shipped data-intensive systems with end-to-end auditability, regulatory compliance, and interoperability with existing partner ecosystems.

**Hashgreen Labs** Sep 2022 – Aug 2023
*Co-founder & CTO* *Boston, MA / Taipei, Taiwan*

Built and led a 20-person engineering organization, raised $1MM+ in capital, and shipped client-facing data and transaction systems while establishing hiring, delivery, and product operating cadence from zero.

- Raised over $1MM USD in capital and grew the team from 0 to 20 employees.
- Built hiring, onboarding, and performance review systems to support a high-performing engineering organization.
- Secured and delivered high-impact projects for clients including The World Bank and Chia Network.
- Led development of DeFi protocols (DEX and AMM), becoming the first team on the Chia blockchain to launch these primitives.

**International Academia of Biomedical Innovation Technology** 2021 – Present
*Consultant & Convener* *Taipei, Taiwan*

Supported a 50+ member biomedical innovation community through project mentorship, dental AI research collaboration, and large-scale technology outreach events.

- Helped grow a community of 50+ members while supporting 8+ academic projects and 15+ publications.
- Led a dental AI research team in collaboration with National Taiwan University Hospital.
- Co-organized demo events with 100+ participants to showcase biomedical technologies and educate the broader community.

**Massachusetts Institute of Technology** Sep 2018 – Jun 2022
*Research Assistant* *Cambridge, MA*

Led research on deployable medical AI, computer vision, and learning under data constraints, producing 20+ publications, patents, and theses with translational partners across industry and healthcare.

- Published 14 conference articles, 5 journal articles, one U.S. patent, and two degree theses.
- Served as a peer reviewer for 8 conferences and journals, reviewing 34 articles.
- Received media coverage from CBS, Engadget, TechCrunch, and MIT News.
- Collaborated with Google, Takeda Pharmaceutical, and Beth Israel Deaconess Medical Center.

- Awarded scholarships totaling 4 years in duration.

**WorldQuant** Jun 2021 – Aug 2021

*Data Science Intern* *Taipei, Taiwan*

Improved an existing quantitative strategy by +1.5% market correlation through targeted modeling and optimization work.

**Digital Drift** Mar 2016 – Mar 2020

*Lead Machine Learning Scientist* *Taipei, Taiwan*

Built and deployed deep learning systems and backend APIs that moved computer vision features from experimentation into production mobile experiences.

- Built deep neural networks for cuisine recognition and matching using TensorFlow on multi-GPU machines.
- Designed and maintained backend APIs to serve model predictions in production settings.

**Google** Jun 2020 – Sep 2020

*Software Engineer Intern* *Remote*

Contributed production software and experimentation tooling that improved reliability and iteration speed for applied machine learning workflows.

**Google** Jun 2019 – Mar 2020

*Researcher* *Seattle, WA / Cambridge, MA*

Built large-scale ML experimentation infrastructure spanning 1,000+ GPUs and 1M+ machine hours, and published 2 peer-reviewed papers in deep federated learning with open-source outputs for broader adoption.

- Conducted foundational research in deep federated learning, contributing to early growth of the field.
- Published 2 peer-reviewed papers and presented findings at top conferences and workshops.
- Developed a scalable experimentation library running on thousands of machines and 1,000+ GPUs (over 1M machine hours).
- Open-sourced research code and datasets for broader community use.

**Brigham and Women's Hospital** Mar 2019 – Feb 2020

*Research Trainee* *Boston, MA*

Contributed hospital-facing medical-imaging ML work on risk prediction and deployment-oriented evaluation using real patient cohorts and radiology workflows.

- Supported model development and validation analysis for clinical imaging applications tied to operational hospital use cases.

**Beth Israel Deaconess Medical Center** Nov 2019 – Oct 2020

*Research Trainee* *Boston, MA*

Published impactful clinical AI research and deployed a PACS-integrated screening workflow across 1,000+ patients while managing shared deep-learning compute for a 10+ member research unit.

- Published three impactful research articles on critical medical topics, including liver tumor assessment, COVID-19 clinical risk prediction, and automatic pancreatic cancer assessment.
- Developed and implemented an AI-driven patient screening pipeline in the hospital's PACS workflow, screening over 1,000 patients and improving disease diagnosis accuracy.
- Managed the computation resources for a research unit of 10+ members, streamlining deep learning processes and maximizing resource sharing.

**Academia Sinica** Feb 2014 – May 2016

*Student Researcher* *Taipei City, Taiwan*

Researched heterogeneous, unsupervised, and semi-supervised domain adaptation for visual classification and published three peer-reviewed papers presented at top conferences and workshops.

- Researched heterogeneous, unsupervised, and semi-supervised domain adaptation for visual classification.
- Published 3 peer-reviewed academic papers and presented findings at top conferences and workshops.

**Olympiad Tutoring Community** Sep 2011 – Jun 2015
*Private Tutor* *Taipei, Taiwan*

Mentored advanced physics students from fundamentals through competition-level performance, including pathways to Taiwan's national Olympiad representation.

- Mentored students who became Taiwan national representatives in the International Physics Olympiad (IPhO).

## *Leadership Experiences*

**Federation of Taiwanese Student Associations in New England (FT-SANE)** 2019 – 2020
*Activities Officer* *Boston, MA*

**MIT Taiwanese Student Association** 2018 – 2019
*President* *Boston, MA*

- Led coordination of 20+ events for a diverse community of over 100 members.
- Organized 10+ career workshops and helped plan 10+ national holiday celebrations.

## *Awards & Honors*

**Departmental Valedictorian (First Place at Graduation)** 2016
*Department of Electrical Engineering, National Taiwan University* *Taipei, Taiwan*

- Placed 1st among 200 classmates across all courses.

**Silver Medal** 2015
*Altera Innovate Asia FPGA Design Competition* *Taipei, Taiwan*

- Placed 2nd among 20 international teams.
- Designed a product named EZBud with integrated algorithms that modulate music based on user sporting statistics.

**Gold Medal / Overall Winner** 2011
*42nd International Physics Olympiad (IPhO)* *Bangkok, Thailand*

- Represented Taiwan and placed 1st in theory, experiment, and combined rankings among 400+ participants from 80+ countries.

**Gold Medal / Experiment Winner** 2011
*12th Asian Physics Olympiad (APhO)* *Tel Aviv, Israel*

- Placed 1st in the experiment section among 100+ participants.

**Gold Medal** 2010
*41st International Physics Olympiad (IPhO)* *Zagreb, Croatia*

**Gold Medal / Experiment Winner** 2010
*11th Asian Physics Olympiad (APhO)* *Taipei, Taiwan*

- Placed 1st in the experiment section among 100+ participants.

**Gold Medal** 2008
*5th International Junior Science Olympiad (IJSO)* *Changwon, Korea*

**Performance Frontier in Freediving: An Exploratory Analysis** 2026
*AIDA Instructor Thesis*
TMH Hsu, J Chang

**USE: Uncertainty Structure Estimation for Robust Semi-Supervised Learning** 2026
*arXiv Preprint arXiv:2603.00404*
Tsao-Lun Chen, Chen-Lin Liu, **Tzu-Ming Harry Hsu**, Tai-Hsien Wu, Ching-Chang Fu, Hsuan-Yi Eric Chou, Shun-Fa Su

**Artificial Intelligence to Assess Dental Findings from Panoramic Radiographs—A Multinational Study** 2025
*arXiv* *Cited by 11*
Yin-Chih Chelsea Wang, Tsao-Lun Chen, Shankeeth Vinayahalingam, Tai-Hsien Wu, Chu Wei Chang, Hsuan Hao Chang, Hung-Jen Wei, Mu-Hsiung Chen, Ching-Chang Ko, David Anssari Moin, Bram van Ginneken, Tong Xi, Hsiao-Cheng Tsai, Min-Huey Chen, **Tzu-Ming Harry Hsu**, Hye Chou

**Intra-Oral Scan Segmentation Using Deep Learning** 2023
*BMC Oral Health* *Cited by 48*
Shankeeth Vinayahalingam, Steven Kempers, Julian Schoep, **Tzu-Ming Harry Hsu**, David Anssari Moin, Bram van Ginneken, Tabea Flugge, Marcel Hanisch, Tong Xi

**Positional Assessment of Lower Third Molar and Mandibular Canal Using Explainable Artificial Intelligence** 2023
*Journal of Dentistry* *Cited by 31*
Steven Kempers, Pieter van Lierop, **Tzu-Ming Harry Hsu**, David Anssari Moin, Stefaan Berge, Hossein Ghaeminia, Tong Xi, Shankeeth Vinayahalingam

**Emulating Clinical Diagnostic Reasoning for Jaw Cysts with Machine Learning** 2022
*Diagnostics* *Cited by 27*
Balazs Feher, Ulrike Kuchler, Falk Schwendicke, Lisa Schneider, Jose Eduardo Cejudo Grano de Oro, Tong Xi, Shankeeth Vinayahalingam, **Tzu-Ming Harry Hsu**, Janet Brinz, Akhilanand Chaurasia

**Methods and Apparatus for Radio Frequency Sensing in Diverse Environments** 2022
*US Patent 11,308,291* *Cited by 18*
Unsoo Ha, Junshan Leng, Alaa Khaddaj, Yunfei Ma, **Tzu Ming Hsu**, Zexuan Zhong, Fadel Adib

**Effective Modeling in Medical Imaging with Constrained Data** 2022
*PhD Thesis, Massachusetts Institute of Technology*
**Tzu-Ming Harry Hsu**

**Artificial Intelligence to Assess Body Composition on Routine Abdominal CT Scans and Predict Mortality in Pancreatic Cancer – A Recipe for Your Local Application** 2021
*European Journal of Radiology* *Cited by 62*
**Tzu-Ming Harry Hsu**, Khoschy Schawkat, Seth J Berkowitz, Jesse L Wei, Alina Makoyeva, Kaila Legare, Corinne DeCicco, S Nicolas Paez, Jim SH Wu, Peter Szolovits

**Visceral Adiposity and Severe COVID-19 Disease: Application of an Artificial Intelligence Algorithm to Improve Clinical Risk Prediction** 2021
*Open Forum Infectious Diseases* *Cited by 23*
Alexander Goehler, **Tzu-Ming Harry Hsu**, Jacqueline A Seiglie, Mark J Siedner, Janet Lo, Virginia Triant, John Hsu, Andrea Foulkes, Ingrid Bassett, Ramin Khorasani, Deborah J Wexler, Peter Szolovits, James B Meigs, Jennifer Manne-Goehler

**Adversarial Contrastive Pre-Training for Protein Sequences** 2021
*arXiv Preprint arXiv:2102.00466* *Cited by 9*

Matthew McDermott, Brendan Yap, **Harry Hsu**, Di Jin, Peter Szolovits

**DeepOPG: Improving Orthopantomogram Finding Summarization with Weak Supervision** 2021
*Medical Image Computing and Computer Assisted Intervention (MICCAI 2021)* *Cited by 8*

**Tzu-Ming Harry Hsu**, Yin-Chih Chelsea Wang

**Methods and Apparatus for Radio Frequency Sensing in Diverse Environments** 2020
*US Patent 10,872,209* *Cited by 18*

Unsoo Ha, Junshan Leng, Alaa Khaddaj, Yunfei Ma, **Tzu Ming Hsu**, Zexuan Zhong, Fadel Adib

**Three-Dimensional Neural Network to Automatically Assess Liver Tumor Burden Change on Consecutive Liver MRIs** 2020
*Journal of the American College of Radiology* *Cited by 23*

Alexander Goehler, **Tzu-Ming Harry Hsu**, Ronilda Lacson, Isha Gujrathi, Raein Hashemi, Grzegorz Chlebus, Peter Szolovits, Ramin Khorasani

**Chexpert++: Approximating the Chexpert Labeler for Speed, Differentiability, and Probabilistic Output** 2020
*Machine Learning for Healthcare Conference (MLHC 2020)* *Cited by 48*

Matthew BA McDermott, **Tzu Ming Harry Hsu**, Wei-Hung Weng, Marzyeh Ghassemi, Peter Szolovits

**Automatic Longitudinal Assessment of Tumor Responses** 2020
*PhD Thesis, Massachusetts Institute of Technology* *Cited by 3*

**Tzu-Ming Harry Hsu**

**Baselines for Chest X-Ray Report Generation** 2020
*Machine Learning for Health (ML4H) Workshop, NeurIPS 2020* *Cited by 97*

William Boag, **Tzu-Ming Harry Hsu**, Matthew McDermott, Gabriela Berner, Emily Alesentzer, Peter Szolovits

**Federated Visual Classification with Real-World Data Distribution** 2020
*European Conference on Computer Vision (ECCV 2020)* *Cited by 289*

**Tzu-Ming Harry Hsu**, Hang Qi, Matthew Brown

**Measuring the Effects of Non-Identical Data Distribution for Federated Visual Classification** 2019
*arXiv Preprint arXiv:1909.06335* *Cited by 2148*

**Tzu-Ming Harry Hsu**, Hang Qi, Matthew Brown

**Transfer Neural Trees: Semi-Supervised Heterogeneous Domain Adaptation and Beyond** 2019
*IEEE Transactions on Image Processing* *Cited by 22*

Wei-Yu Chen, **Tzu-Ming Harry Hsu**, Yao-Hung Hubert Tsai, Ming-Syan Chen, Yu-Chiang Frank Wang

**Clinically Accurate Chest X-Ray Report Generation** 2019
*Machine Learning for Healthcare Conference (MLHC 2019)* *Cited by 418*

Guanxiong Liu, **Tzu-Ming Harry Hsu**, Matthew McDermott, Willie Boag, Wei-Hung Weng, Peter Szolovits, Marzyeh Ghassemi

**Unsupervised Multimodal Representation Learning Across Medical Images and Reports** 2018
*Machine Learning for Health (ML4H) Workshop, NeurIPS 2018* *Cited by 53*

**Tzu-Ming Harry Hsu**, Wei-Hung Weng, Willie Boag, Matthew McDermott, Peter Szolovits

**Learning Food Quality and Safety from Wireless Stickers** 2018
*Proceedings of the 17th ACM Workshop on Hot Topics in Networks (HotNets 2018)* *Cited by 98*

Unsoo Ha, Yunfei Ma, Zexuan Zhong, **Tzu-Ming Hsu**, Fadel Adib

**3D-Aware Scene Manipulation via Inverse Graphics** 2018
*Advances in Neural Information Processing Systems (NeurIPS 2018)* *Cited by 324*

Shunyu Yao, **Tzu Ming Hsu**, Jun-Yan Zhu, Jiajun Wu, Antonio Torralba, Bill Freeman, Josh Tenenbaum

**Transfer Neural Trees for Heterogeneous Domain Adaptation** 2016
*European Conference on Computer Vision (ECCV 2016)* *Cited by 83*

Wei-Yu Chen, **Tzu-Ming Harry Hsu**, Yao-Hung Hubert Tsai, Yu-Chiang Frank Wang, Ming-Syan Chen

**Connecting the Dots Without Clues: Unsupervised Domain Adaptation for Cross-Domain Visual Classification** 2015
*International Conference on Image Processing (ICIP 2015)* *Cited by 2*

Wei-Yu Chen, **Tzu-Ming Harry Hsu**, Cheng-An Hou, Yi-Ren Yeh, Yu-Chiang Frank Wang

**Unsupervised Domain Adaptation with Imbalanced Cross-Domain Data** 2015
*International Conference on Computer Vision (ICCV 2015)* *Cited by 90*

**Tzu Ming Harry Hsu**, Wei Yu Chen, Cheng-An Hou, Yao-Hung Hubert Tsai, Yi-Ren Yeh, Yu-Chiang Frank Wang